Diploma in Electronics / Electronics & Communication Engineering / Biomedical Engineering • Semester 3

Electronic Circuits Lab

Hands-on course for BJT/FET characteristics, CE amplifier, emitter follower, tuned amplifier, oscillators, multivibrators, Schmitt trigger and UJT relaxation oscillator.

Course Code: 3047 | **Category: Program Core** | **Credits: 1.5** | **L:T:P = 0:0:3** | **Lab / Practical**

COURSE OVERVIEW

Electronic Circuits Lab (3047)

This is a standalone digital textbook, revision guide, workbook and answer guide prepared from the uploaded official syllabus PDF.

Student-friendly summary

Hands-on course for BJT/FET characteristics, CE amplifier, emitter follower, tuned amplifier, oscillators, multivibrators, Schmitt trigger and UJT relaxation oscillator.

Study tip: □□□ module-□□ separate □□□ □□□□□□□□. Definitions, diagrams, formulas/procedures, and expected answers English technical terms □□□□□□□□□□ practise □□□□□□□.

Course Metadata

ITEM	DETAILS
Course Code	3047
Base Course Code	3047
Course Title	Electronic Circuits Lab
Programme / Department	Diploma in Electronics / Electronics & Communication Engineering / Biomedical Engineering

ITEM	DETAILS
Semester	3
Course Category	Program Core
Credits	1.5
L:T:P	0:0:3
Periods / semester	45
Subject Type	Lab / Practical

Course Objectives

- Provide hands-on experience with basic electronic devices and circuits.
- Study characteristics of electronic devices.
- Impart design procedures of electronic circuits.
- Develop troubleshooting skills.
- Familiarize usage of instruments and measuring equipment.

Prerequisite Knowledge

- Basic Mathematics - Mathematics I, II
- Experiments with electric circuits and electronic components - Fundamentals of Electrical & Electronics Lab
- Characteristics of diodes and transistors - Basic Electronics

Course Outcomes

CO	DESCRIPTION	HOURS	LEVEL
CO1	Experiment with input and output characteristics of BJT, FET and UJT.	9	Applying
CO2	Design transistor-based amplifier circuits.	12	Applying
CO3	Design low-frequency and high-frequency oscillator circuits.	9	Applying
CO4	Design transistor-based switching circuits.	9	Applying

Module Map

1. **Module 1:** BJT, FET and UJT characteristics (9 h)
2. **Module 2:** Transistor amplifier circuits (12 h)
3. **Module 3:** Oscillator circuits (9 h)

4. **Module 4:** Transistor switching circuits (9 h)

How to study this subject

- Read the aim and apparatus before lab.
- Draw the circuit/setup diagram before wiring.
- Write procedure and observation table in advance.
- After experiment, write result and precautions immediately.
- Practise viva questions module-wise.

Exam preparation method

- Memorise lab record format: aim, apparatus, theory, diagram, procedure, observation, result.
- Practise neat wiring/setup explanation.
- Prepare safety precautions and viva answers.
- Know how to calculate/read final result from observations.

Quick Navigation Cards

EXPERIMENT BANK

Experiment Bank - Electronic Circuits Lab

Quick reference cards for revision. Use this only after reading the module explanation.

Record format

Aim -> Apparatus -> Theory -> Circuit -> Procedure -> Observation -> Graph -> Result

Meaning: Complete lab record flow.

Where used: All experiments.

Common mistake: Forgetting graph scale.

Amplifier gain

$$A_v = V_{out}/V_{in}$$

Meaning: Voltage gain calculation.

Where used: CE amplifier/emitter follower.

Common mistake: Using peak-to-peak and RMS inconsistently.

Bandwidth

$$BW = f_H - f_L$$

Meaning: Frequency response width.

Where used: CE/tuned amplifier experiments.

Common mistake: Not marking 3 dB points.

Oscillator frequency

$$f = 1/T$$

Meaning: Measure from waveform period.

Where used: RC/Hartley/crystal/UJT oscillators.

Common mistake: Wrong time/div reading.

Schmitt trigger levels

UTP and LTP

Meaning: Upper/lower threshold points.

Where used: Schmitt trigger experiment.

Common mistake: Not showing hysteresis loop.

MODULE 1 • 9 HOURS

BJT, FET and UJT characteristics

Build circuits and plot input/output characteristics of BJT and FET.

Learning outcome

Build circuits and plot input/output characteristics of BJT and FET.

Malayalam support: □□□□□ meaning □□□□□□□□□□□□□, □□□□□□ English technical terms □□□□□□□□□□ answer □□□□□□.

Important terms

- BJT CE input characteristics
- BJT CE output characteristics
- FET drain characteristics in CS configuration
- Observation table and graph plotting
- Device region identification

Official module outcome table

SYLLABUS CODE	MODULE OUTCOME	HOURS	LEVEL
M1.01	Build circuit and plot input characteristics of BJT in CE configuration	3	Applying
M1.02	Construct circuit and plot output characteristics of BJT in CE configuration	3	Applying
M1.03	Construct circuit and plot output/drain characteristics of FET in CS configuration	3	Applying

Topic-wise explanation

BJT CE input characteristics: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

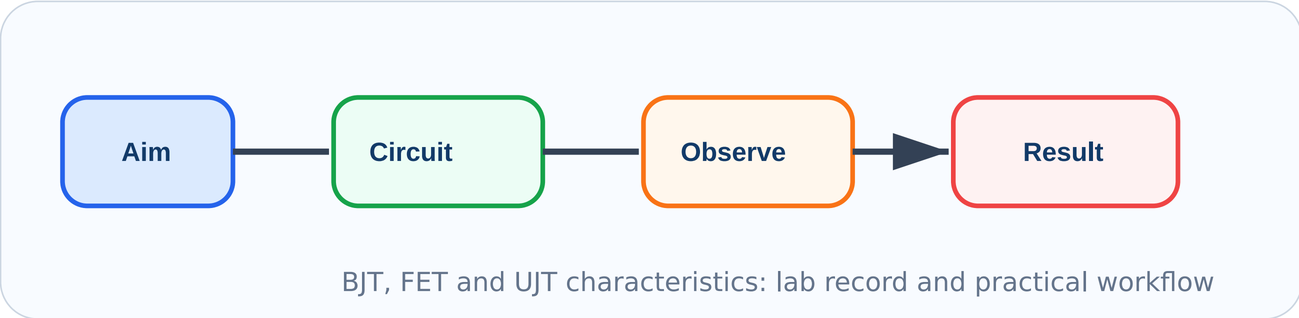
BJT CE output characteristics: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

FET drain characteristics in CS configuration: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Observation table and graph plotting: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Device region identification: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Visual block / diagram idea



Use this type of labelled diagram/block in answers when applicable.

Worked / practice example

Practical example: For amplifier frequency response, apply constant input, vary frequency, record output voltage, calculate gain, plot gain vs frequency and identify bandwidth.

Common mistakes

- Wiring the circuit without switching off the supply.
- Using wrong IC pin number or wrong instrument range.
- Not drawing waveform axes and scale clearly.
- Writing result without observation or calculation.
- Ignoring component polarity and maximum ratings.

Expected Questions

M1-Q1 Write the aim and apparatus required for an experiment based on BJT, FET and UJT characteristics.

M1-Q2 Explain the step-by-step procedure for BJT, FET and UJT characteristics.

M1-Q3 Draw or describe the required circuit/setup and observation table for BJT, FET and UJT characteristics.

M1-Q4 List important precautions and safety points while doing BJT, FET and UJT characteristics.

M1-Q5 Write viva questions with answers for BJT, FET and UJT characteristics.

M1-Q6 How will you write the final result and verify correctness in BJT, FET and UJT characteristics?

Module checklist

✓ Aim and apparatus are clear.

✓ Circuit/setup is neat and labelled.

✓ Procedure is in correct order.

✓ Observation table and result are complete.

✓ Precautions and viva points are prepared.

MODULE 2 • 12 HOURS

Transistor amplifier circuits

Design CE amplifier, emitter follower and tuned amplifier; observe phase, gain and frequency response.

Learning outcome

Design CE amplifier, emitter follower and tuned amplifier; observe phase, gain and frequency response.

Malayalam support: □□□□□ meaning □□□□□□□□□□□□□, □□□□□□ English technical terms □□□□□□□□□□ answer □□□□□□.

Important terms

- Single stage CE amplifier with potential divider bias
- Phase difference between input and output
- Mid-band gain measurement
- Frequency response and bandwidth
- Emitter follower circuit
- Input/output waveform plotting
- Single stage tuned amplifier
- Peak gain measurement

Official module outcome table

SYLLABUS CODE	MODULE OUTCOME	HOURS	LEVEL
M2.01	Design CE amplifier and measure phase difference and mid-band gain	3	Applying
M2.02	Plot frequency response and measure bandwidth of CE amplifier	3	Applying
M2.03	Build emitter follower and measure gain/waveforms	3	Applying
M2.04	Construct tuned amplifier, measure peak gain, response and bandwidth	3	Applying

Topic-wise explanation

Single stage CE amplifier with potential divider bias: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Phase difference between input and output: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Mid-band gain measurement: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Frequency response and bandwidth: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

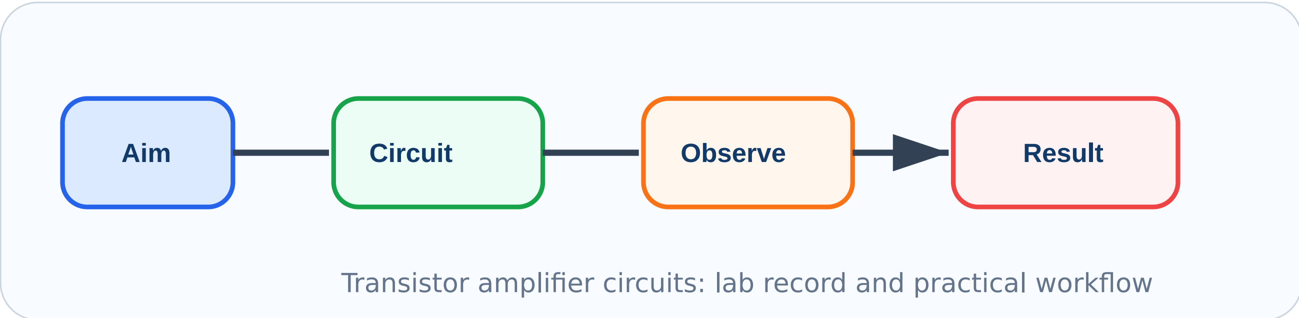
Emitter follower circuit: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Input/output waveform plotting: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Single stage tuned amplifier: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Peak gain measurement: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Visual block / diagram idea



Use this type of labelled diagram/block in answers when applicable.

Worked / practice example

Practical example: For amplifier frequency response, apply constant input, vary frequency, record output voltage, calculate gain, plot gain vs frequency and identify bandwidth.

Common mistakes

- Wiring the circuit without switching off the supply.
- Using wrong IC pin number or wrong instrument range.
- Not drawing waveform axes and scale clearly.
- Writing result without observation or calculation.
- Ignoring component polarity and maximum ratings.

Expected Questions

M2-Q1 Write the aim and apparatus required for an experiment based on Transistor amplifier circuits.

M2-Q2 Explain the step-by-step procedure for Transistor amplifier circuits.

M2-Q3 Draw or describe the required circuit/setup and observation table for Transistor amplifier circuits.

M2-Q4 List important precautions and safety points while doing Transistor amplifier circuits.

M2-Q5 Write viva questions with answers for Transistor amplifier circuits.

M2-Q6 How will you write the final result and verify correctness in Transistor amplifier circuits?

Module checklist

- ✓ Aim and apparatus are clear.
- ✓ Circuit/setup is neat and labelled.
- ✓ Procedure is in correct order.
- ✓ Observation table and result are complete.
- ✓ Precautions and viva points are prepared.

MODULE 3 • 9 HOURS

Oscillator circuits

Design RC phase shift, Hartley and crystal oscillator circuits and measure frequency.

Learning outcome

Design RC phase shift, Hartley and crystal oscillator circuits and measure frequency.

Malayalam support: □□□□□ meaning □□□□□□□□□□□□□, □□□□□□ English technical terms □□□□□□□□□□ answer □□□□□□.

Important terms

RC phase shift oscillator | **Output waveform plotting** | **Frequency measurement from CRO/DSO** | **Hartley oscillator** | **Crystal oscillator circuit**

Comparison of oscillator stability

Official module outcome table

SYLLABUS CODE	MODULE OUTCOME	HOURS	LEVEL
M3.01	Design RC phase shift oscillator and measure frequency	3	Applying
M3.02	Design Hartley oscillator and measure frequency	3	Applying
M3.03	Construct crystal oscillator and measure frequency	3	Applying

Topic-wise explanation

RC phase shift oscillator: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Output waveform plotting: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

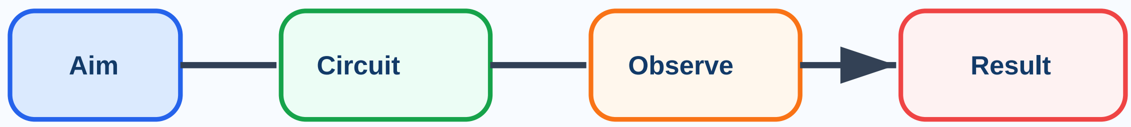
Frequency measurement from CRO/DSO: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Hartley oscillator: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Crystal oscillator circuit: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Comparison of oscillator stability: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Visual block / diagram idea



Oscillator circuits: lab record and practical workflow

Use this type of labelled diagram/block in answers when applicable.

Worked / practice example

Practical example: For amplifier frequency response, apply constant input, vary frequency, record output voltage, calculate gain, plot gain vs frequency and identify bandwidth.

Common mistakes

- Wiring the circuit without switching off the supply.
- Using wrong IC pin number or wrong instrument range.
- Not drawing waveform axes and scale clearly.
- Writing result without observation or calculation.
- Ignoring component polarity and maximum ratings.

Expected Questions

M3-Q1 Write the aim and apparatus required for an experiment based on Oscillator circuits.

M3-Q2 Explain the step-by-step procedure for Oscillator circuits.

M3-Q3 Draw or describe the required circuit/setup and observation table for Oscillator circuits.

M3-Q4 List important precautions and safety points while doing Oscillator circuits.

M3-Q5 Write viva questions with answers for Oscillator circuits.

M3-Q6 How will you write the final result and verify correctness in Oscillator circuits?

Module checklist

- ✓ Aim and apparatus are clear.
- ✓ Circuit/setup is neat and labelled.
- ✓ Procedure is in correct order.
- ✓ Observation table and result are complete.
- ✓ Precautions and viva points are prepared.

MODULE 4 • 9 HOURS

Transistor switching circuits

Design astable multivibrator, Schmitt trigger and UJT relaxation oscillator circuits.

Learning outcome

Design astable multivibrator, Schmitt trigger and UJT relaxation oscillator circuits.

Malayalam support: □□□□□ meaning □□□□□□□□□□□□□, □□□□□□ English technical terms □□□□□□□□□□ answer □□□□□□.

Important terms

- **Astable multivibrator using transistor**
- **Collector and base waveforms**
- **Frequency of oscillation measurement**
- **Schmitt trigger using BJT**
- **UTP and LTP voltage measurement**
- **UJT relaxation oscillator**
- **Emitter and base waveform plotting**

Official module outcome table

SYLLABUS CODE	MODULE OUTCOME	HOURS	LEVEL
M4.01	Design astable multivibrator and plot collector/base waveforms	3	Applying
M4.02	Construct Schmitt trigger and measure UTP/LTP	3	Applying
M4.03	Design UJT relaxation oscillator and plot waveforms	3	Applying

Topic-wise explanation

Astable multivibrator using transistor: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Collector and base waveforms: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Frequency of oscillation measurement: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

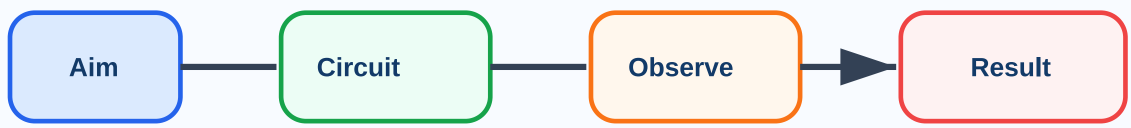
Schmitt trigger using BJT: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

UTP and LTP voltage measurement: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

UJT relaxation oscillator: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Emitter and base waveform plotting: Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

Visual block / diagram idea



Transistor switching circuits: lab record and practical workflow

Use this type of labelled diagram/block in answers when applicable.

Worked / practice example

Practical example: For amplifier frequency response, apply constant input, vary frequency, record output voltage, calculate gain, plot gain vs frequency and identify bandwidth.

Common mistakes

- Wiring the circuit without switching off the supply.
- Using wrong IC pin number or wrong instrument range.
- Not drawing waveform axes and scale clearly.
- Writing result without observation or calculation.
- Ignoring component polarity and maximum ratings.

Expected Questions

M4-Q1 Write the aim and apparatus required for an experiment based on Transistor switching circuits.

M4-Q2 Explain the step-by-step procedure for Transistor switching circuits.

M4-Q3 Draw or describe the required circuit/setup and observation table for Transistor switching circuits.

M4-Q4 List important precautions and safety points while doing Transistor switching circuits.

M4-Q5 Write viva questions with answers for Transistor switching circuits.

M4-Q6 How will you write the final result and verify correctness in Transistor switching circuits?

Module checklist

✓ Aim and apparatus are clear.

✓ Circuit/setup is neat and labelled.

✓ Procedure is in correct order.

✓ Observation table and result are complete.

✓ Precautions and viva points are prepared.

QUICK REVISION

One-page Revision Summary

Use this page in the last 24 hours before exam or lab evaluation.

Module-wise must-know points

1. **Module 1 - BJT, FET and UJT characteristics:** Prepare circuit, procedure, observation table and graph for BJT/FET characteristics.
2. **Module 2 - Transistor amplifier circuits:** Know amplifier circuits, waveform relation, gain, response curve and bandwidth method.
3. **Module 3 - Oscillator circuits:** Prepare oscillator circuits, expected waveform and frequency calculation from time period.
4. **Module 4 - Transistor switching circuits:** Know switching circuit setup, threshold measurement, waveform plotting and frequency calculation.

Important definitions / keywords

BJT CE input characteristics | BJT CE output characteristics | FET drain characteristics | CE amplifier | emitter follower | tuned amplifier | frequency response | RC phase shift oscillator | Hartley oscillator | crystal oscillator | astable multivibrator | Schmitt trigger | UJT relaxation oscillator

Important formulas / key points

- Aim -> Apparatus -> Theory -> Circuit -> Procedure -> Observation -> Graph -> Result
- $A_v = V_{out}/V_{in}$
- $BW = f_H - f_L$
- $f = 1/T$
- UTP and LTP

Common exam mistakes

- Wiring the circuit without switching off the supply.
- Using wrong IC pin number or wrong instrument range.
- Not drawing waveform axes and scale clearly.
- Writing result without observation or calculation.
- Ignoring component polarity and maximum ratings.

Last-minute checklist

✓ Memorise lab record format: aim, apparatus, theory, diagram, procedure, observation, result.

✓ Practise neat wiring/setup explanation.

✓ Prepare safety precautions and viva answers.

✓ Know how to calculate/read final result from observations.

Diagram / table list

- BJT CE input/output characteristic circuit
- FET drain characteristic circuit
- CE amplifier circuit
- Frequency response plot
- Emitter follower circuit
- Single tuned amplifier circuit
- RC phase shift oscillator
- Hartley oscillator
- Crystal oscillator
- Astable multivibrator waveforms
- Schmitt trigger waveform
- UJT relaxation oscillator waveform

MASTER ANSWER KEY

Master Answer Key - matched with Expected Questions

Every answer below matches the module expected questions exactly. Attempt first, then verify here.

M1-Q1 - Write the aim and apparatus required for an experiment based on BJT, FET and UJT characteristics.

Aim: Perform the experiment related to BJT, FET and UJT characteristics and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M1-Q2 - Explain the step-by-step procedure for BJT, FET and UJT characteristics.

Aim: Perform the experiment related to BJT, FET and UJT characteristics and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M1-Q3 - Draw or describe the required circuit/setup and observation table for BJT, FET and UJT characteristics.

Aim: Perform the experiment related to BJT, FET and UJT characteristics and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M1-Q4 - List important precautions and safety points while doing BJT, FET and UJT characteristics.

Aim: Perform the experiment related to BJT, FET and UJT characteristics and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M1-Q5 - Write viva questions with answers for BJT, FET and UJT characteristics.

Aim: Perform the experiment related to BJT, FET and UJT characteristics and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M1-Q6 - How will you write the final result and verify correctness in BJT, FET and UJT characteristics?

Aim: Perform the experiment related to BJT, FET and UJT characteristics and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M2-Q1 - Write the aim and apparatus required for an experiment based on Transistor amplifier circuits.

Aim: Perform the experiment related to Transistor amplifier circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M2-Q2 - Explain the step-by-step procedure for Transistor amplifier circuits.

Aim: Perform the experiment related to Transistor amplifier circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M2-Q3 - Draw or describe the required circuit/setup and observation table for Transistor amplifier circuits.

Aim: Perform the experiment related to Transistor amplifier circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M2-Q4 - List important precautions and safety points while doing Transistor amplifier circuits.

Aim: Perform the experiment related to Transistor amplifier circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M2-Q5 - Write viva questions with answers for Transistor amplifier circuits.

Aim: Perform the experiment related to Transistor amplifier circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M2-Q6 - How will you write the final result and verify correctness in Transistor amplifier circuits?

Aim: Perform the experiment related to Transistor amplifier circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M3-Q1 - Write the aim and apparatus required for an experiment based on Oscillator circuits.

Aim: Perform the experiment related to Oscillator circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M3-Q2 - Explain the step-by-step procedure for Oscillator circuits.

Aim: Perform the experiment related to Oscillator circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M3-Q3 - Draw or describe the required circuit/setup and observation table for Oscillator circuits.

Aim: Perform the experiment related to Oscillator circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M3-Q4 - List important precautions and safety points while doing Oscillator circuits.

Aim: Perform the experiment related to Oscillator circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M3-Q5 - Write viva questions with answers for Oscillator circuits.

Aim: Perform the experiment related to Oscillator circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M3-Q6 - How will you write the final result and verify correctness in Oscillator circuits?

Aim: Perform the experiment related to Oscillator circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M4-Q1 - Write the aim and apparatus required for an experiment based on Transistor switching circuits.

Aim: Perform the experiment related to Transistor switching circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M4-Q2 - Explain the step-by-step procedure for Transistor switching circuits.

Aim: Perform the experiment related to Transistor switching circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M4-Q3 - Draw or describe the required circuit/setup and observation table for Transistor switching circuits.

Aim: Perform the experiment related to Transistor switching circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M4-Q4 - List important precautions and safety points while doing Transistor switching circuits.

Aim: Perform the experiment related to Transistor switching circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M4-Q5 - Write viva questions with answers for Transistor switching circuits.

Aim: Perform the experiment related to Transistor switching circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

M4-Q6 - How will you write the final result and verify correctness in Transistor switching circuits?

Aim: Perform the experiment related to Transistor switching circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.